

8. Lie algebras

8.1. The Jacobi identity. The matrix commutator $[x, y] = xy - yx$ obviously satisfies the identity

$$[[x, y], z] + [[y, z], x] + [[z, x], y] = 0$$

called the **Jacobi identity**. Thus it is satisfied for any Lie subgroup of $GL_n(\mathbb{K})$.

Proposition 8.1. *The Jacobi identity holds for any Lie group G .*

Proof. Let $\mathfrak{g} = T_1G$. The Jacobi identity is equivalent to adx being a derivation of the commutator:

$$\text{adx}([y, z]) = [\text{adx}(y), z] + [y, \text{adx}(z)], \quad x, y, z \in \mathfrak{g}.$$

To show that it is indeed a derivation, let $g(t) = \exp(tx)$, then

$$\text{Ad}_{g(t)}([y, z]) = [\text{Ad}_{g(t)}(y), \text{Ad}_{g(t)}(z)].$$

The desired identity is then obtained by differentiating this equality by t at $t = 0$ and using the Leibniz rule and Proposition 7.11(iv). $\square$

Corollary 8.2. *We have $\text{ad}[x, y] = [\text{adx}, \text{ady}]$.*

Proof. This is also equivalent to the Jacobi identity. $\square$

Proposition 8.3. *For $x \in \mathfrak{g}$ one has $\exp(\text{adx}) = \text{Ad}_{\exp(x)} \in GL(\mathfrak{g})$.*

Proof. We will show that $\exp(t\text{adx}) = \text{Ad}_{\exp(tx)}$ for $t \in \mathbb{R}$. Let $\gamma_1(t) = \exp(t\text{adx})$ and $\gamma_2(t) = \text{Ad}_{\exp(tx)}$. Then γ_1, γ_2 both satisfy the differential equation $\gamma'(t) = \gamma(t)\text{adx}$ and equal 1 at $t = 0$. Thus $\gamma_1 = \gamma_2$. $\square$

8.2. Lie algebras.

Definition 8.4. A **Lie algebra** over a field $\mathbf{k}$ is a vector space $\mathfrak{g}$ over $\mathbf{k}$ equipped with bilinear operation $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$, called the **commutator** or **(Lie) bracket** which satisfies the following identities:

- (i) $[x, x] = 0$ for all $x \in \mathfrak{g}$;
- (ii) the Jacobi identity: $[[x, y], z] + [[y, z], x] + [[z, x], y] = 0$.

A **(homo)morphism of Lie algebras** is a linear map between Lie algebras that preserves the commutator.

Remark 8.5. If $\mathbf{k}$ has characteristic $\neq 2$ then the condition $[x, x] = 0$ is equivalent to skew-symmetry $[x, y] = -[y, x]$, but in characteristic 2 it is stronger.

Example 8.6. Any subspace of $\mathfrak{gl}_n(\mathbf{k})$ closed under $[x, y] := xy - yx$ is a Lie algebra.

Example 8.7. The map $\text{ad} : \mathfrak{g} \rightarrow \text{End}(\mathfrak{g})$ is a morphism of Lie algebras.

Thus we have

Theorem 8.8. *If G is a $\mathbb{K}$ -Lie group (for $\mathbb{K} = \mathbb{R}, \mathbb{C}$) then $\mathfrak{g} := T_1G$ has a natural structure of a Lie algebra over $\mathbb{K}$. Moreover, if $\phi : G \rightarrow K$ is a morphism of Lie groups then $\phi_* : T_1G \rightarrow T_1K$ is a morphism of Lie algebras.*

We will denote the Lie algebra $\mathfrak{g} = T_1G$ by $\text{Lie}G$ or $\text{Lie}(G)$ and call it the **Lie algebra of G** . We see that the assignment $G \mapsto \text{Lie}G$ is a functor from the category of Lie groups to the category of Lie algebras. Thus we have a map $\text{Hom}(G, K) \rightarrow \text{Hom}(\text{Lie}G, \text{Lie}K)$, which is injective if G is connected.

Motivated by Proposition 7.11(v), a Lie algebra $\mathfrak{g}$ is said to be **commutative** or **abelian** if $[x, y] = 0$ for all $x, y \in \mathfrak{g}$.

8.3. Lie subalgebras and ideals. A **Lie subalgebra** of a Lie algebra $\mathfrak{g}$ is a subspace $\mathfrak{h} \subset \mathfrak{g}$ closed under the commutator. It is called a **Lie ideal** if moreover $[\mathfrak{g}, \mathfrak{h}] \subset \mathfrak{h}$.

Proposition 8.9. *Let $H \subset G$ be a Lie subgroup. Then:*

- (i) *$\text{Lie}H \subset \text{Lie}G$ is a Lie subalgebra;*
- (ii) *If H is normal then $\text{Lie}H$ is a Lie ideal in $\text{Lie}G$;*
- (iii) *If G, H are connected and $\text{Lie}H \subset \text{Lie}G$ is a Lie ideal then H is normal in G .*

Proof. (i) If $x, y \in \mathfrak{h}$ then $\exp(tx), \exp(sy) \in H$, so by Proposition 7.11(iv)

$$[x, y] = \lim_{t, s \rightarrow 0} \frac{\log(\exp(tx) \exp(sy) \exp(-tx) \exp(-sy))}{ts} \in \mathfrak{h}.$$

(ii) We have $ghg^{-1} \in H$ for $g \in G$ and $h \in H$. Thus, taking $h = \exp(sy)$, $y \in \mathfrak{h}$ and taking the derivative in s at zero, we get $\text{Ad}_g(y) \in \mathfrak{h}$. Now taking $g = \exp(tx)$, $x \in \mathfrak{g}$ and taking the derivative in t at zero, by Proposition 7.11(iv) we get $[x, y] \in \mathfrak{h}$, i.e., $\mathfrak{h}$ is a Lie ideal.

(iii) If $x \in \mathfrak{g}$, $y \in \mathfrak{h}$ are small then

$$\begin{aligned} \exp(x) \exp(y) \exp(x)^{-1} &= \\ \exp(\text{Ad}_{\exp(x)} y) &= \exp(\exp(\text{ad} x) y) = \exp\left(\sum_{n=0}^{\infty} \frac{(\text{ad} x)^n y}{n!}\right) \in H \end{aligned}$$

since $\sum_{n=0}^{\infty} \frac{(\text{adx})^n y}{n!} \in \mathfrak{h}$. So G acting on itself by conjugation maps a small neighborhood of 1 in H into H (as G is generated by its neighborhood of 1 by Proposition 3.15, since it is connected). But H is also connected, so is generated by its neighborhood of 1, again by Proposition 3.15. Hence H is normal. $\square$

8.4. The Lie algebra of vector fields. Recall that a vector field on a manifold X is a compatible family of derivations $\mathbf{v} : O(U) \rightarrow O(U)$ for open subsets $U \subset X$.

Proposition 8.10. *If $\mathbf{v}, \mathbf{w}$ are derivations of an algebra A then so is $[\mathbf{v}, \mathbf{w}] := \mathbf{v}\mathbf{w} - \mathbf{w}\mathbf{v}$.*

Proof. We have

$$\begin{aligned} (\mathbf{v}\mathbf{w} - \mathbf{w}\mathbf{v})(ab) &= \mathbf{v}(\mathbf{w}(a)b + a\mathbf{w}(b)) - \mathbf{w}(\mathbf{v}(a)b + a\mathbf{v}(b)) = \\ &\quad \mathbf{v}\mathbf{w}(a)b + \mathbf{w}(a)\mathbf{v}(b) + \mathbf{v}(a)\mathbf{w}(b) + a\mathbf{v}\mathbf{w}(b) \\ &\quad - \mathbf{w}\mathbf{v}(a)b - \mathbf{v}(a)\mathbf{w}(b) - \mathbf{w}(a)\mathbf{v}(b) - a\mathbf{w}\mathbf{v}(b) = \\ &\quad (\mathbf{v}\mathbf{w} - \mathbf{w}\mathbf{v})(a)b + a(\mathbf{v}\mathbf{w} - \mathbf{w}\mathbf{v})(b). \end{aligned}$$

$\square$

Thus, the space $\text{Vect}(X)$ of vector fields on X is a Lie algebra under the operation

$$\mathbf{v}, \mathbf{w} \mapsto [\mathbf{v}, \mathbf{w}],$$

called the **Lie bracket of vector fields**.⁷

In local coordinates we have

$$\mathbf{v} = \sum_i v_i \frac{\partial}{\partial x_i}, \quad \mathbf{w} = \sum_j w_j \frac{\partial}{\partial x_j},$$

so

$$[\mathbf{v}, \mathbf{w}] = \sum_i \left(\sum_j (v_j \frac{\partial w_i}{\partial x_j} - w_j \frac{\partial v_i}{\partial x_j}) \right) \frac{\partial}{\partial x_i}.$$

This implies that if vector fields $\mathbf{v}, \mathbf{w}$ are tangent to a k -dimensional submanifold $Y \subset X$ then so is their Lie bracket $[\mathbf{v}, \mathbf{w}]$. Indeed, in local coordinates Y is given by equations $x_{k+1} = \dots = x_n = 0$, and in such coordinates a vector field is tangent to Y iff it does not contain terms with $\frac{\partial}{\partial x_j}$ for $j > k$.

⁷Note that this Lie algebra is infinite dimensional for all real manifolds and many (but not all) complex manifolds of positive dimension.

Exercise 8.11. Let $U \subset \mathbb{R}^n$ be an open subset, $\mathbf{v}, \mathbf{w} \in \text{Vect}(U)$ and g_t, h_t be the associated flows, defined in a neighborhood of every point of U for small t . Show that for any $\mathbf{x} \in U$

$$\lim_{t,s \rightarrow 0} \frac{g_t h_s g_t^{-1} h_s^{-1}(\mathbf{x}) - \mathbf{x}}{ts} = [\mathbf{v}, \mathbf{w}](\mathbf{x}).$$

Now let G be a Lie group and $\text{Vect}_L(G), \text{Vect}_R(G) \subset \text{Vect}(G)$ be the subspaces of left and right invariant vector fields.

Proposition 8.12. $\text{Vect}_L(G), \text{Vect}_R(G) \subset \text{Vect}(G)$ are Lie subalgebras which are both canonically isomorphic to $\mathfrak{g} = \text{Lie}G$.

Proof. The first statement is obvious, so we prove only the second statement. Let $\mathbf{x}, \mathbf{y} \in \text{Vect}_L(G)$. Then $\mathbf{x} = \mathbf{L}_x, \mathbf{y} = \mathbf{L}_y$ for $x = \mathbf{x}(1), y = \mathbf{y}(1) \in \mathfrak{g}$, where $\mathbf{L}_z$ denotes the vector field on G obtained by left translations of $z \in \mathfrak{g}$. Then $[\mathbf{L}_x, \mathbf{L}_y] = \mathbf{L}_z$, where $z = [\mathbf{L}_x, \mathbf{L}_y](1)$. So let us compute z .

Let f be a regular function on a neighborhood of $1 \in G$. We have shown that for $u \in \mathfrak{g}$

$$(\mathbf{L}_u f)(g) = \frac{d}{dt} \Big|_{t=0} f(g \exp(tu)).$$

Thus,

$$\begin{aligned} z(f) &= x(\mathbf{L}_y f) - y(\mathbf{L}_x f) = x\left(\frac{\partial}{\partial s} \Big|_{s=0} f(\bullet \exp(sy))\right) - y\left(\frac{\partial}{\partial t} \Big|_{t=0} f(\bullet \exp(tx))\right) = \\ &= \frac{\partial}{\partial t} \Big|_{t=0} \frac{\partial}{\partial s} \Big|_{s=0} f(\exp(tx) \exp(sy)) - \frac{\partial}{\partial s} \Big|_{s=0} \frac{\partial}{\partial t} \Big|_{t=0} f(\exp(sy) \exp(tx)) = \\ &= \frac{\partial^2}{\partial t \partial s} \Big|_{t=s=0} (F(tx + sy + \tfrac{1}{2}ts[x, y] + \dots) - F(tx + sy - \tfrac{1}{2}ts[x, y] + \dots)), \end{aligned}$$

where $F(u) := f(\exp(u))$. It is easy to see by using Taylor expansion that this expression equals to $[x, y](f)$. Thus $z = [x, y]$, i.e., the map $\mathfrak{g} \rightarrow \text{Vect}_L(G)$ given by $x \mapsto \mathbf{L}_x$ is a Lie algebra isomorphism. Similarly, the map $\mathfrak{g} \rightarrow \text{Vect}_R(G)$ given by $x \mapsto -\mathbf{R}_x$ is a Lie algebra isomorphism, as claimed. $\square$

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